

2018 HKDSE Physics

Paper 1B — Structured Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Section B. Answer ALL questions. Parts marked with * involve knowledge of the extension component.

Question 1

Figure 1.1 shows a domestic water heater. Tap water entering the heater is heated when it passes through the coil duct and exits as hot water at a certain temperature.

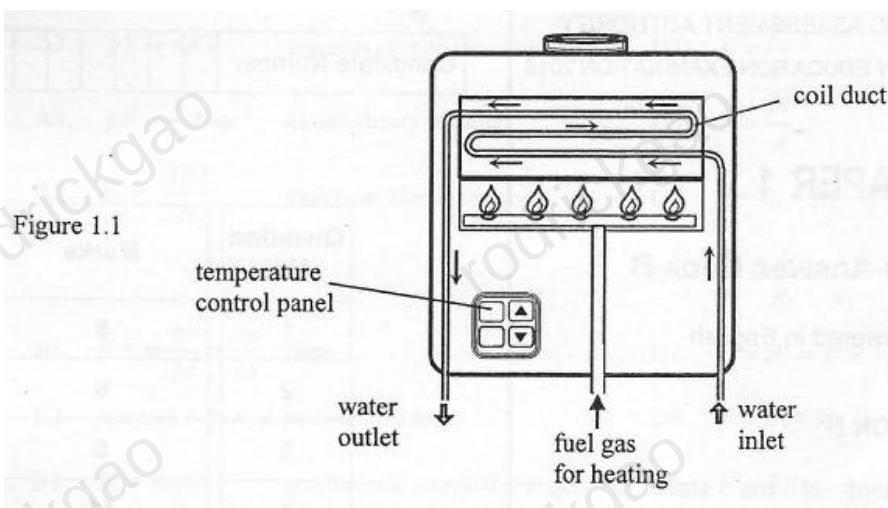


Figure 1.1

On a certain day in winter, the temperature of tap water is $15\text{ }^{\circ}\text{C}$. When the heater is in use, it delivers 6 kg of hot water at a temperature of $50\text{ }^{\circ}\text{C}$ in 1 minute . Assume that there is no heat exchange between the heater and the surroundings.

Given: specific heat capacity of water = $4200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

- Estimate the power supplied to the tap water by the heater. (3 marks)
- Assuming the power estimated in (a) remains unchanged, determine the flow rate of tap water into the heater, in kg per minute, such that hot water at a temperature of $40\text{ }^{\circ}\text{C}$ is delivered by the heater. (2 marks)

Official marking scheme

Paper 1 Section B		
Solution	Marks	Remarks
1. (a) Amount of energy required $E = mc\Delta T$ $= 6 \times 4200 \times (50 - 15)$ $= 882000 \text{ J (or } 882 \text{ kJ)}$	1M	
Power $P = \frac{E}{t} = \frac{882000}{60}$ $= 14700 \text{ W (or } 14.7 \text{ kW)}$	1M	
	1A	
	3	
(b) Let m kg per minute be the water flow rate $mc\Delta T = Pt$ $m(4200)(40 - 15) = 14700 \times 60$ $m = 8.4 \text{ (kg min}^{-1} \text{ or kg)}$	1M	Or $\frac{m}{t} \propto \frac{1}{\Delta T}$
	1A	$\therefore \frac{m}{6} = \frac{50-15}{40-15}$
	2	
2. (a) $n = \frac{pV}{RT} \propto pV$ (for constant T)		

(a) $E = mc\Delta T = 6 \times 4200 \times (50 - 15) = 8.82 \times 10^5 \text{ J}$. Therefore $P = E/t = 8.82 \times 10^5 / 60 = 1.47 \times 10^4 \text{ W} = 14.7 \text{ kW}$.

(b) Let the flow be $m \text{ kg min}^{-1}$. $m(4200)(40 - 15) = (1.47 \times 10^4)(60)$, giving $m = 8.4 \text{ kg min}^{-1}$.

Question 2 *

Two vessels, A and B, of volumes $3V$ and $2V$ respectively are connected by a narrow tube with a tap as shown in Figure 2.1. Initially the tap is closed and both vessels are at the same temperature. Vessel A contains helium gas at a pressure p while vessel B contains 0.8 mol helium gas at a pressure $2p$. Assume that helium gas can be taken as an ideal gas.

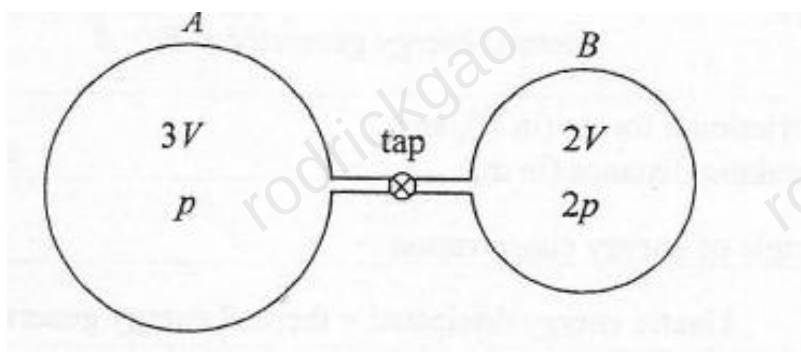


Figure 2.1

- Deduce the amount of helium gas (in mol) in vessel A. (2 marks)
- Now the tap is open and a steady state is reached. Assume that the temperature remains unchanged.
 - Find the gas pressure inside the vessels in terms of p . (2 marks)
 - Account for the pressure change of the gas in vessel A using kinetic theory. (2 marks)

Official marking scheme

$n = \frac{pV}{RT} \propto pV$ (for constant T)	1M
$\frac{n_A}{n_B} = \frac{(p)(3V)}{(2p)(2V)}$	
$n_A = 0.75 \times 0.8 \text{ mol}$	1A
$= 0.6 \text{ (mol)}$	
Or $(2p)(2V) = 0.8RT$	1M
$pV = 0.2RT$	
$p(3V) = nRT$	1A
$n = 3 \times 0.2 = 0.6 \text{ (mol)}$	
	2
(b) (i) $n = n_A + n_B$	
$p'(3V + 2V) = p(3V) + (2p)(2V)$	1M
$p' = 1.4p$	1A
Or $p'(3V + 2V) = (0.6 + 0.8)RT$	1M
$p'(5V) = 1.4RT$	
$p' = 1.4p$	1A
	2
(ii) When the tap is open, number of gas molecules in vessel A increases due to the (net) flow of molecules from B to A, the frequency of collision of gas molecules with the vessel's wall increases, pressure increases.	1A
	1A
	2

(a) At the same temperature, $n \propto pV$. Thus $n_A/n_B = (p \cdot 3V)/(2p \cdot 2V) = 3/4$, so $n_A = 0.60 \text{ mol}$.

(b) (i) Conservation of gas gives $pf(3V+2V) = p(3V)+2p(2V)$, hence $pf = 1.4p$.

(ii) Molecules flow net from B to A; the number density and wall-collision frequency in A increase, so its pressure rises.

Question 3

Read the following passage about braking in vehicles and answer the questions that follow.

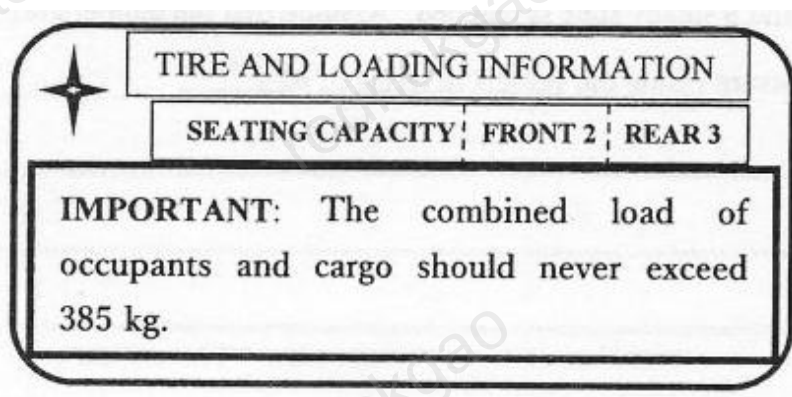
Mechanical braking is the most common braking method in vehicles. In this method, wheels of vehicles are fitted with brake pads which apply frictional forces that inhibit the motion of the wheels. Frictional braking results in a conversion of the vehicle's kinetic energy to thermal energy which then dissipates to the atmosphere in the form of waste heat.

The thermal energy (in J) generated during braking is approximately given by: thermal energy generated = $F_r \times d$, where F_r is the frictional force (in N), and d is the braking distance (in m).

Applying the principle of energy conservation: kinetic energy dissipated = thermal energy generated; $\frac{1}{2}mv^2 = F_r \times d$.

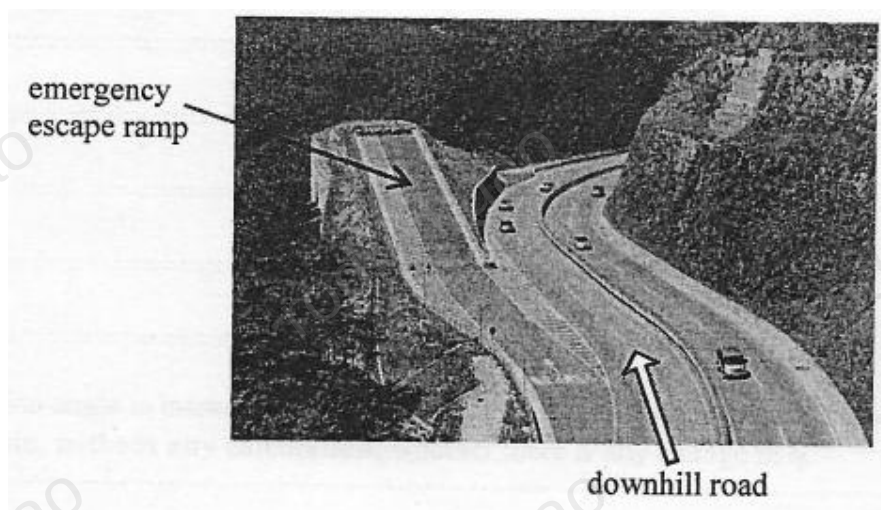
From this equation it can be seen that increasing the velocity v or mass m of an object means the applied frictional force must be increased in order to bring the object to a stop within the same distance.

(a) Referring to the passage, explain why there should be a maximum load limit for a vehicle (see the vehicle label example below). (2 marks)



Vehicle loading label

- (b) (i) It is known that the brake pads would provide much smaller frictional forces or even brake failure would occur (i.e. without any friction provided) when reaching a temperature of over a few hundred degrees Celsius. Explain why it is not recommended for a vehicle driver to apply the brakes continuously during a long downhill trip. (1 mark)
- (ii) The figure shows an emergency escape ramp (slanting upwards) built for stopping vehicles with brake failure resulting from the situation described in (b)(i). If such a ramp makes an angle of 30° with the horizontal and a vehicle with brake failure enters the bottom end of the ramp at a speed of 25 m s^{-1} , estimate how far it will travel along the ramp before it stops. Neglect air resistance and mechanical resistances within the vehicle. ($g = 9.81 \text{ m s}^{-2}$) (2 marks)



Emergency escape ramp

Official marking scheme

Solution	Marks	Remarks
3. (a) If the maximum load is exceeded, the braking distance will increase if the friction provided remains the same. Vehicles would not be able to stop in time in case of emergency (thus dangerous).	1A	
Or A larger friction is required in braking the vehicle within the same distance, accident may occur if the brakes cannot provide such frictional forces.	1A	
	2	
(b) (i) If the brakes are applied continuously, thermal energy generated will heat up the brake pads / brakes to too high a temperature that the brakes may fail.	1A	
	1	
(ii) Let D be the distance travelled along the ramp. Kinetic energy of vehicle becomes its gravitational potential energy: $\frac{1}{2}m(25)^2 = m(9.81)(D \sin 30^\circ)$ $D = 63.710499 \text{ m}$ $\approx 63.7 \text{ m (62.5 m for } g = 10 \text{ m s}^{-2}\text{)}$	1M	Or work done against vehicle's weight component along the ramp by its kinetic energy: $\frac{1}{2}mv^2 = mg \sin \theta \times D$
Or $v^2 = u^2 + 2as$	1A	mass of vehicle = m Note: $D \sin 30^\circ = 31.8552 \text{ m}$
	1M	

(a) For the same speed, increasing the load increases $\frac{1}{2}mv^2$. If the braking friction is unchanged, a longer stopping distance is required; otherwise the vehicle may fail to stop safely, hence a load limit.

(b) (i) Continuous braking converts energy to heat continuously, raising the pad temperature until friction is greatly reduced or the brakes fail.

(ii) $\frac{1}{2}mv^2 = mgD \sin 30^\circ$, so $D = 25^2 / [2(9.81)\sin 30^\circ] = 63.7 \text{ m}$.

Question 4

Figures 4.1(a) and 4.1(b) show the cross-section of a fixed spring gun fitted with a small cannon ball.

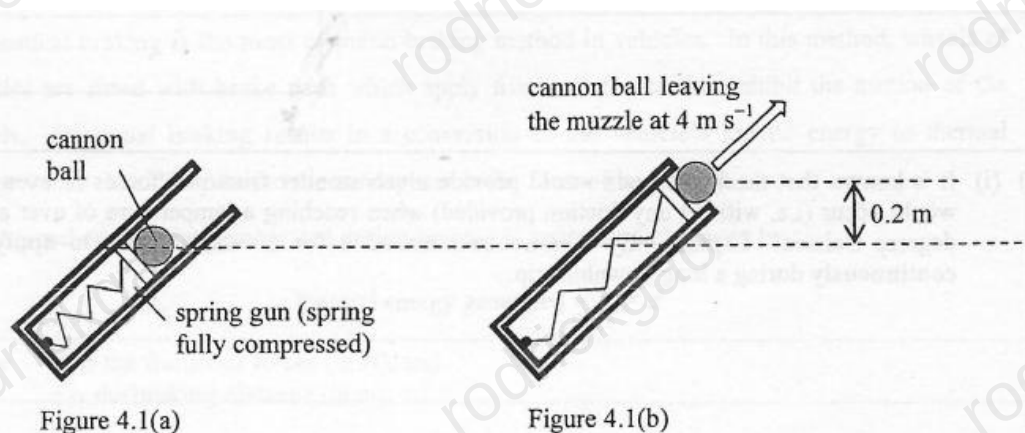


Figure 4.1

The fully compressed light spring (Figure 4.1(a)) is released so that the cannon ball of mass 0.3 kg leaves the muzzle of the gun at a speed of 4 m s^{-1} (Figure 4.1(b)). Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

(a) During the process from the time when the spring is fully compressed till the cannon ball just leaves the muzzle:

- (i) how much energy is transferred from the spring to the cannon ball? (3 marks)
- (ii) explain whether the total momentum of the spring gun and the cannon ball is conserved. (2 marks)
- * (b) The cannon ball is projected at 50° to the horizontal initially and reaches a point horizontally away from the muzzle of the gun at a distance R . Find R and the time of flight t_1 of the cannon ball to that point. (4 marks)
- * (c) If the projection angle is increased to slightly greater than 50° while the initial speed remains unchanged at 4 m s^{-1} , explain, without any calculation, whether there is any change in t_1 . (2 marks)

Official marking scheme

4. (a) (i)	Or $v^2 = u^2 + 2as$ $0^2 = 25^2 + 2(-9.81 \sin 30^\circ)D$ $D \approx 63.7 \text{ m}$	1M	mass of vehicle = m Note: $D \sin 30^\circ = 31.8552 \text{ m}$ (or 31.25 m for $g = 10 \text{ m s}^{-2}$)
		1A	Accept $D = 62.0 \text{ m}$ to 64.0 m
2			
	K.E. + P.E. = $\frac{1}{2}(0.3)(4)^2 + (0.3)(9.81)(0.2)$ $= 2.4 + 0.5586 = 2.9886 \text{ J}$ $\approx 2.99 \text{ J}$ (3.0 J for $g = 10 \text{ m s}^{-2}$)	1M+1M	
		1A	$= 2.4 + 0.6 = 3.0 \text{ J}$ for $g = 10 \text{ m s}^{-2}$
3			
(ii)	As the spring gun is fixed, there is external force acting on the system / the gun, total momentum (of the spring gun and cannon ball) is not conserved.	1M	
		1A	
2			
(b)	Vertical : $s = ut + \frac{1}{2}at^2$ $0 = (4 \sin 50^\circ)t_f - \frac{1}{2}(9.81)t_f^2$ $t_f = 0.624705 \text{ s}$ (0.612836 s for $g = 10 \text{ m s}^{-2}$) $\approx 0.625 \text{ s}$ (0.613 s for $g = 10 \text{ m s}^{-2}$)	1M	Or $\frac{t_f}{2} = \frac{u \sin \theta}{g} = \frac{4 \sin 50^\circ}{9.81}$
		1A	Accept $t_f = 0.61 \text{ s}$ to 0.63 s
	Horizontal : $R = 4 \cos 50^\circ \times t_f = 4 \cos 50^\circ \times 0.625$ $= 1.606210 \text{ m}$ $\approx 1.61 \text{ m}$ (1.57 m for $g = 10 \text{ m s}^{-2}$)	1M	Or $R = u \cos \theta \times \frac{2(u \sin \theta)}{g}$
		1A	Accept $R = 1.56 \text{ m}$ to 1.62 m

(a) (i) Energy transferred = $\Delta K + \Delta U = \frac{1}{2}(0.3)(4^2) + (0.3)(9.81)(0.20) = 2.99 \text{ J}$.

(ii) The fixed gun experiences an external support force, so momentum of the gun–ball system is not conserved.

(b) $t_1 = 2u \sin 50^\circ / g = 0.625 \text{ s}$; $R = u \cos 50^\circ \cdot t_1 = 1.61 \text{ m}$.

(c) For an angle slightly above 50° , $\sin \theta$ is larger, so $t = 2u \sin \theta / g$ increases.

Question 5

Figure 5.1 shows the use of a beam balance to measure the mass of a load. Without the load and the counter-weight, the beam with a scale pan at its left end is balanced and remains horizontal. ($g = 9.81 \text{ m s}^{-2}$)

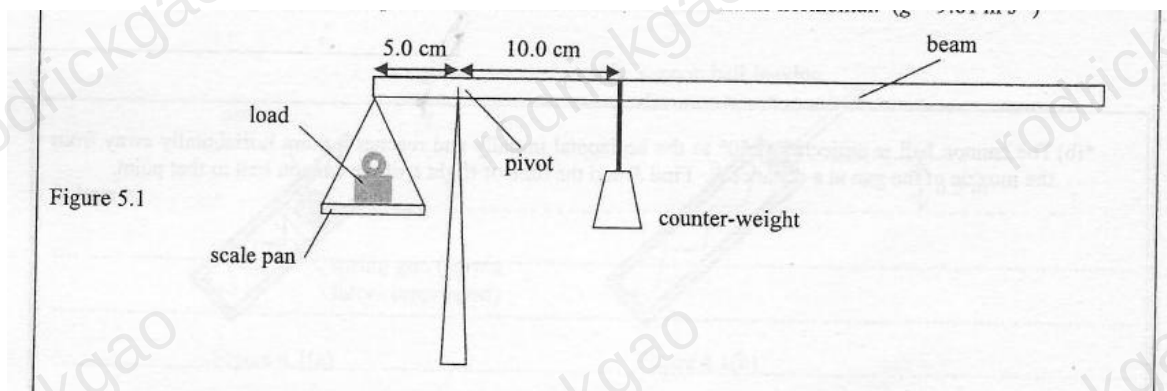


Figure 5.1

- (a) A load is placed on the pan which is 5.0 cm from the pivot. The set-up is in equilibrium when the counter-weight of mass 50 g is 10.0 cm away from the pivot as shown.
- Find the mass of the load. (2 marks)
 - If the reading of the counter-weight's position on the beam carries an uncertainty of ± 0.1 cm, find the maximum error that corresponds to the result found in (a)(i). (2 marks)
- (b) The weight of an identical load is now measured by a spring balance calibrated in newtons. What is the balance's reading? (1 mark)
- (c) The beam balance set-up in Figure 5.1 and the spring balance with the load in (b) are both brought into a lift.
- The measurements are then repeated inside the lift which accelerates upwards uniformly. State the change, if any, in the measurements respectively. (2 marks)
 - A student claims that if the lift falls freely, the beam balance can still be used to measure the mass of the load. Explain whether the claim is correct or not. (2 marks)

Official marking scheme

Solution		Marks	Remarks				
5. (a)	(i) $m(5.0 \text{ cm}) = 50 \text{ g}(10.0 \text{ cm})$ $m = 100 \text{ g}$ or 0.1 kg	1M 1A					
		2					
	(ii) Counter-weight position : $10.0 \text{ cm} \pm 0.1 \text{ cm}$ Percentage error = $100\% \times \left(\frac{0.1}{10.0}\right) = 1\%$ $\therefore m = 101 \text{ g}$ to 99 g i.e. maximum error = $\pm 1 \text{ g}$	1M 1A					
		2					
	(b) Spring balance reading = $mg = (0.1 \text{ kg})(9.81 \text{ N kg}^{-1})$ $= 0.981 \text{ N}$ (1.0 N for $g = 10 \text{ N kg}^{-1}$) $\approx 1.0 \text{ N}$	1M/1A					
		1					
	(c) (i)						
	<table border="1"> <tr> <td>counter-weight position on beam balance</td> <td>spring balance reading</td> </tr> <tr> <td>the same</td> <td>reading increases</td> </tr> </table>	counter-weight position on beam balance	spring balance reading	the same	reading increases	1A+1A	
	counter-weight position on beam balance	spring balance reading					
	the same	reading increases					
		2					
	(ii) The beam balance would fail to work / to measure the mass of the load, as the apparent weight is zero (or weightless), the counter-weight can take any position.	1A 1A					
	2						

(a) (i) Taking moments: $m(5.0 \text{ cm}) = (50 \text{ g})(10.0 \text{ cm})$, so $m = 100 \text{ g}$.

(ii) Fractional uncertainty = $0.1/10.0 = 1\%$, hence maximum mass error = 1 g .

(b) $W = mg = 0.100 \times 9.81 = 0.981 \text{ N}$.

(c) (i) The beam-balance position is unchanged because the same effective gravity multiplies both moments; the spring-balance reading increases.

(ii) In free fall all parts are weightless, so no balancing torque is available and the beam balance cannot measure the mass.

Question 6

- (a) You are given a ray box with a single slit (producing a fine light beam), a cylindrical concave lens, a plastic ruler, a pencil and a piece of paper as shown in Figure 6.1.

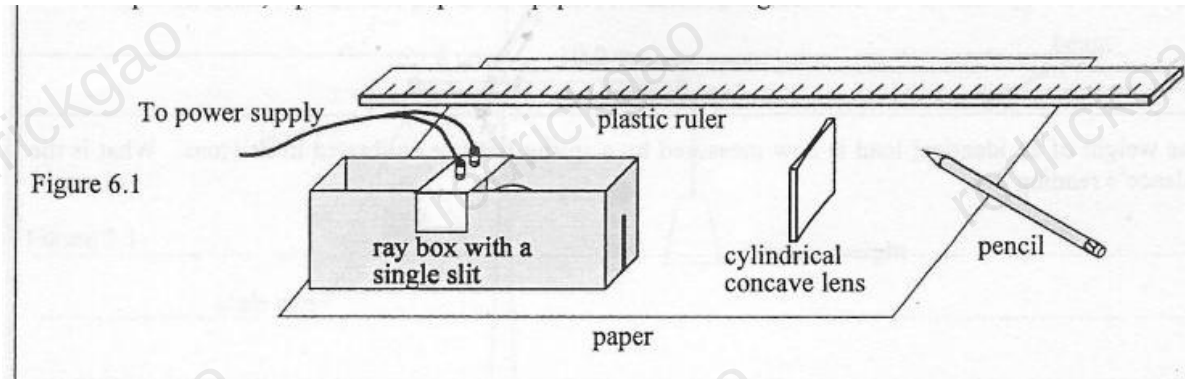
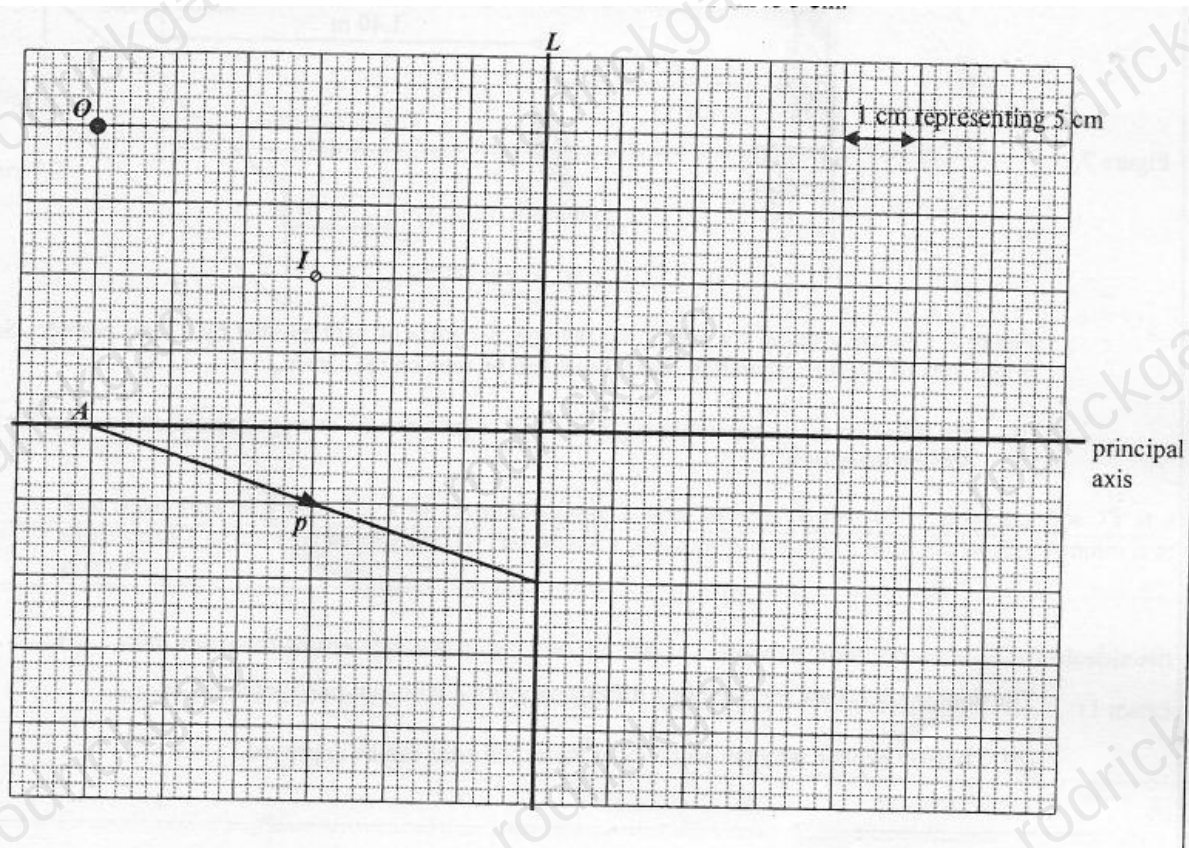


Figure 6.1

Describe how you would use the above apparatus to find the focal length of the lens and state ONE possible source of error in the experiment. (5 marks)

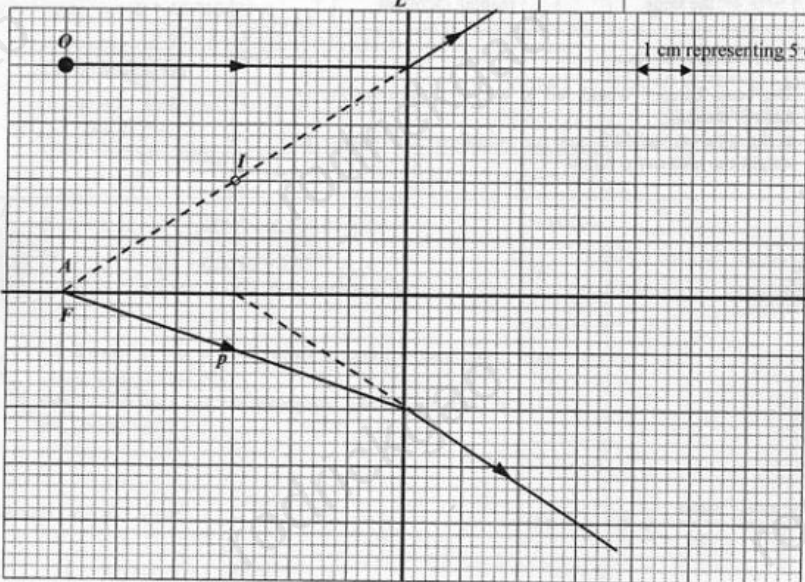
- (b) In the figure below, L represents another cylindrical lens. A vertical pin used as the object is placed at O; the image is formed at I by the lens. The horizontal scale is 1 cm representing 5 cm.



Ray diagram (horizontal scale: 1 cm represents 5 cm)

- (i) What kind of lens is used? Explain. (2 marks)
- (ii) Draw a suitable light ray to locate the principal focus F of lens L . Find its focal length. (2 marks)
- (iii) Complete the path for the ray p from point A . (1 mark)

Official marking scheme

Solution		Marks	Remarks
6. (a)	Place the (cylindrical) lens on the piece of paper and trace its outline, use ray tracing method e.g. - direct and trace a light ray towards the lens. - direct and trace a light ray along / parallel to the principal axis	1A	Or Using lens formula: - use ray tracing method: direct and trace a light ray (not parallel to the principal axis) towards the lens
	Direct another light ray parallel to the first one / principal axis (by shifting the ray box) and trace the path(s) of the (emerging) ray(s) on the paper.	1A	- extend (the path of) the emerging ray backward and locate the point of intersection (with the principal axis)
	Extend (the path of) the emerging rays (backward) and locate the point of intersection (on the focal plane containing F).	1A	- measure the corresponding object distance u and image distance v (on the principal axis)
	Measure the distance from the point of intersection (or F) to the centre of the lens, which gives the focal length of the lens.	1A	- sub. u and v into lens formula to find f
	Source of error: Scale uncertainty/accuracy/precision of the plastic ruler (read to nearest mm). OR Unable to mark the path correctly because of the thickness of the beam of light from the ray box OR The ray(s) is/are not parallel (to the principal axis) OR Any reasonable answer	1A	Accept: Using diagram(s) to supplement / simplify the description
		5	
(b)			
			
(i) L is diverging/concave. - only a diverging lens can produce a (virtual / erect / diminished) image between the object and the lens		1A 1A 2	Accept (30 ± 1) cm
(ii) Focal length = 30 cm Correct ray to find F		1A 1A 2	
(iii) Correct ray p		1A 1	

(a) Trace the lens outline. Send two narrow, parallel rays through the concave lens, trace their incident and emergent paths, and extend the emergent rays backwards to their common virtual focus. The distance from this point to the lens centre is $|f|$. A valid error is ruler resolution, finite beam width or rays not exactly parallel.

(b) (i) The lens is concave/diverging because it forms a virtual, erect, diminished image between object and lens.

- (ii) Use a ray through the optical centre and extend the appropriate refracted ray to locate F; the scale gives $|f| = 30 \text{ cm}$.
- (iii) Complete ray p so its backward extension passes through the virtual image point.

Question 7

- (a) Figure 7.1 shows a set-up for measuring the wavelength λ of light emitted by a laser pointer. Several bright dots of average separation about 2 mm can be seen on the screen.

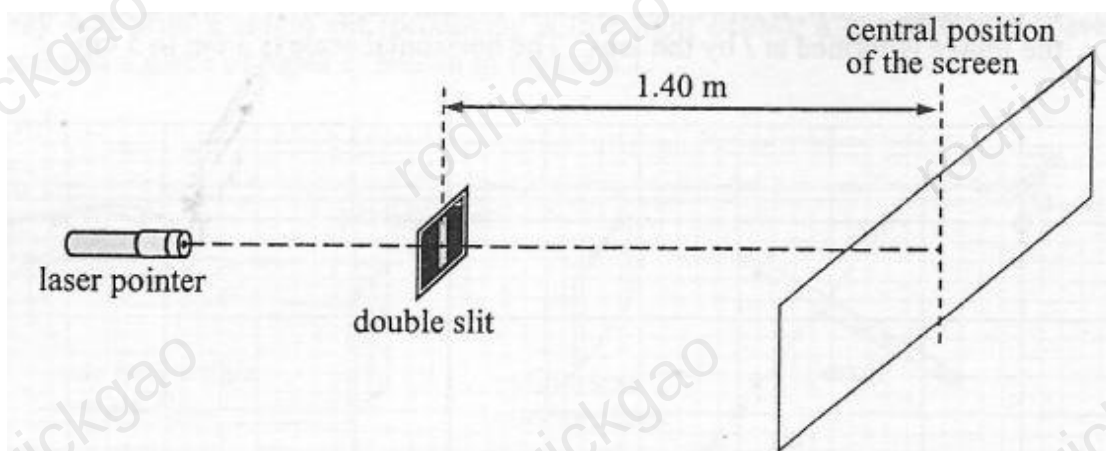


Figure 7.1

- (i) For the same set of apparatus, suggest a way to increase the average separation between the bright dots on the screen. (1 mark)

The double slit is now replaced by a diffraction grating with 400 lines per mm.

- (ii) Briefly explain why the accuracy of the experiment can be improved. (1 mark)
- (iii) Only five bright dots are observed on the screen such that the separation between the 1st and 5th dots is 1.56 m. Find λ . (3 marks)

- (b) To measure the speed of sound in air, a student connects two loudspeakers, A and B, to a signal generator as shown in Figure 7.2. The separation of A and B is 1 m. A microphone is used to pick up the sound along the line XY at a distance of 2 m from the loudspeakers. The central maximum is at point O while the next maximum is at point P.

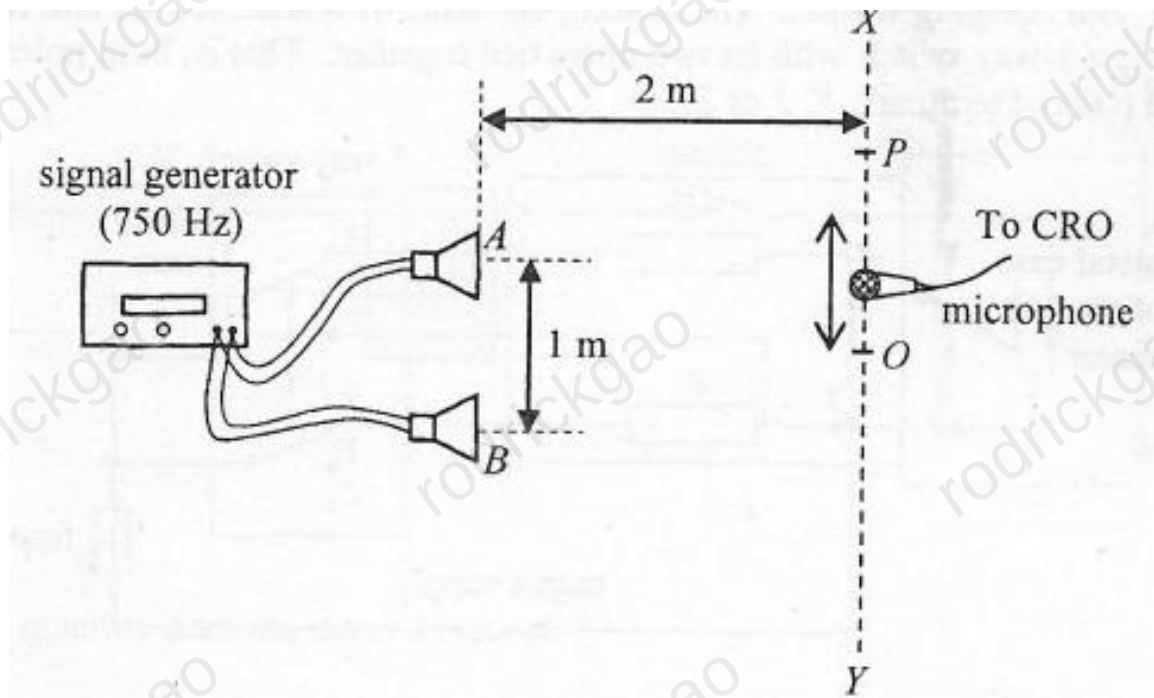
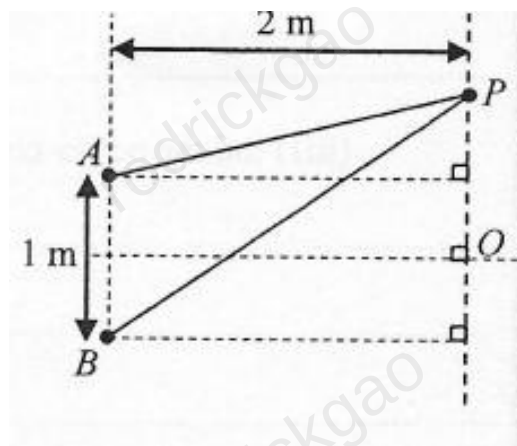


Figure 7.2

- (i) With reference to the above settings, use the fringe separation equation $\Delta y = \lambda D/a$ in double-slit interference to find the wavelength λ of sound is not accurate. Explain briefly. (1 mark)
- (ii) The distance between O and P is found to be 1 m when the signal generator is set at 750 Hz. By considering the path difference $PB - PA$, use the results of the experiment to find the speed of sound in air. (3 marks)



Geometry for paths PA and PB

Official marking scheme

Solution		Marks	Remarks
7. (a)	(i) Increase the separation between the double slit and the screen, D .	1A	
		1	
	(ii) The separation of the bright dots on the screen becomes larger, thus the percentage error in its measurement is smaller.	1A	
		1	
	(iii) The angular position of the 2 nd order bright dot $\theta = \tan^{-1}\left(\frac{1.56/2}{1.40}\right) = 29.124053^\circ$	1M	
	Grating spacing $d = \frac{10^{-3}}{400} = 2.5 \times 10^{-6} \text{ m}$	1M	
	Applying $d \sin \theta = n\lambda$, Wavelength $\lambda = \frac{2.5 \times 10^{-6} \times \sin 29.12^\circ}{2}$ $= 6.08378 \times 10^{-7} \text{ m}$ $\approx 6.08 \times 10^{-7} \text{ m} (= 608 \text{ nm})$	1A	
		3	
	(b) (i) The equation can only be applied for - $\lambda \ll a$ (i.e. wavelength $\ll$ separation of the two sources), OR λ is much smaller than a - $a \ll D$ (i.e. separation of the two sources $\ll$ separation of the sources and detector), OR a is much smaller than D	Any ONE 1A	
	Or Using the fringe separation equation to find the wavelength of sound is not accurate for - λ is comparable to/ NOT much smaller than a - a is NOT much smaller than D	Any ONE 1A	
(ii)	For the 1 st order maximum, wavelength $\lambda = \text{path difference } PB - PA$ $= \sqrt{(1+0.5)^2 + 2^2} - \sqrt{(1-0.5)^2 + 2^2}$ $= 2.5 - 2.06155281 = 0.43844719 \text{ m} \approx 0.438 \text{ m}$	1M	Note : the order of magnitude of the wavelength of sound is about 10^{-1} m
	speed of sound: $v = f\lambda = 750 \times 0.4384$ $= 328.835 \text{ m s}^{-1}$ $\approx 329 \text{ m s}^{-1}$	1M	
		1A	
		3	

(a) (i) Increase the slit–screen distance D .

(ii) The grating gives more widely separated bright spots, reducing percentage uncertainty.

(iii) For the second order, $\theta = \tan^{-1}[(1.56/2)/1.40]$. With $d = 1/(400 \text{ mm}^{-1}) = 2.5 \times 10^{-6} \text{ m}$, $\lambda = d \sin \theta / 2 = 6.08 \times 10^{-7} \text{ m} = 608 \text{ nm}$.(b) (i) The approximation $\Delta y = \lambda D/a$ requires $\lambda \ll a$ and $a \ll D$; here neither condition is strong.(ii) $\lambda = PB - PA = \sqrt{(1.5^2 + 2^2)} - \sqrt{(0.5^2 + 2^2)} = 0.438 \text{ m}$, so $v = f\lambda = 750 \times 0.438 = 329 \text{ m s}^{-1}$.

Question 8

- (a) Figure 8.1 shows the schematic diagram of an electric heater which can operate in two modes, namely, ‘heating’ and ‘keeping warm’. The heating elements of resistances $4R$ and R are connected to the mains supply via a 3-way switch with its two poles tied together. That is, both poles can be connected to one of the three pairs of terminals X, Y or Z.

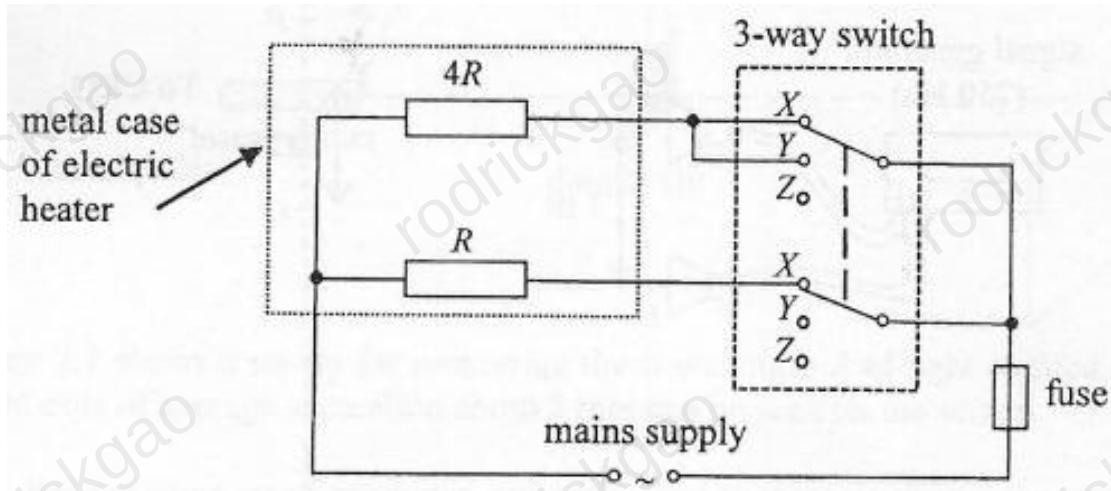


Figure 8.1

- (i) To which pair of terminals, X, Y or Z, should the switch connect when the heater is in ‘heating’ mode? (1 mark)

The power consumed by the heater in ‘heating’ mode is 800 W.

- (ii) Calculate the current drawn from the 220 V mains supply when the heater is in ‘heating’ mode. (2 marks)
- (iii) Find the power consumed by the heater in the mode of ‘keeping warm’. (3 marks)

- (b) Figure 8.2 shows a simplified domestic circuit connected to an electrical appliance via a fuse, a meter M, a residual current circuit breaker (RCCB) and a switch.

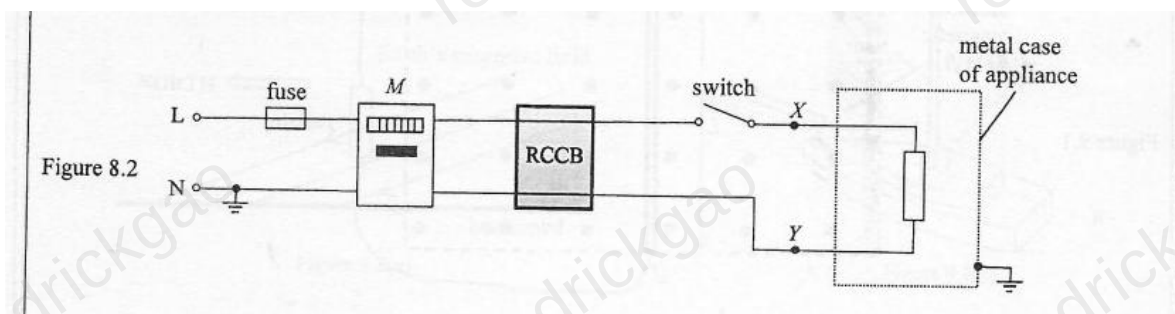


Figure 8.2

- (i) What physical quantity does the meter M record? (1 mark)
- (ii) An RCCB is a kind of safety device that cuts off the supply automatically whenever there is a small difference between the currents in the live (L) and neutral (N) wires. State, in each of the following situations, which device(s) will respond (i.e. the fuse blows and/or the RCCB cuts off the supply).
- (1) A short circuit occurs between points X and Y. (1 mark)

(2) A short circuit occurs between point Y and the metal case of the appliance. (1 mark)

Official marking scheme

Solution		Marks	Remarks
8. (a)	(i) To terminals X	1A	
		1	
	(ii) $P = IV$ $800 \text{ W} = I(220 \text{ V})$ $I = 3.636364 \text{ A}$ $\approx 3.64 \text{ A}$	1M 1A	
		2	
	(iii) $800 = \frac{V^2}{R} + \frac{V^2}{4R} = \frac{5V^2}{4R}$ $P_{\text{keep warm}} = \frac{V^2}{4R}$ $= 800\left(\frac{1}{5}\right) = 160 \text{ W}$	1M 1M 1A	
	Alternative (I) $800 = \frac{V^2}{R} + \frac{V^2}{4R} = \frac{5V^2}{4R}$ $R = 75.625 \Omega$ $P_{\text{keep warm}} = \frac{V^2}{4R}$ $= \frac{220^2}{4(75.625)}$ $= 160 \text{ W}$	1M 1M 1A	
	Alternative (II) Power $\propto \frac{1}{\text{resistance}}$ $\frac{P_{\text{keep warm}}}{P_{\text{heating}}} = \frac{R_{\text{eq}}}{4R}$ $P_{\text{keep warm}} = P_{\text{heating}} \left(\left(\frac{1}{R} + \frac{1}{4R} \right)^{-1} / 4R \right)$ $= 160 \text{ W}$	1M 1M 1A	
		3	
	(b) (i) Electrical energy (consumed)	1A 1	
	(ii) (1) only the fuse blows (2) only the RCCB cuts off the supply	1A 1A	
		2	

(a) (i) Connect to terminals X.

(ii) $I = P/V = 800/220 = 3.64 \text{ A}$.

(iii) Heating uses R and 4R in parallel, whereas keeping warm uses 4R only. Thus $P_{\text{warm}}/P_{\text{heat}} = [V^2/(4R)]/[V^2/R + V^2/(4R)] = 1/5$, so $P_{\text{warm}} = 160 \text{ W}$.

(b) (i) M records electrical energy consumed.

(ii) An X–Y short circuit makes only the fuse blow; a Y–case fault makes only the RCCB cut off the supply.

Question 9

- (a) Figure 9.1 shows a metal rod PQ of length L moving with constant velocity v across a uniform magnetic field B pointing out of the paper. An e.m.f. is induced across rod PQ as it cuts the field lines. When the rod is connected to a resistor outside the field, a current I flows in the circuit.

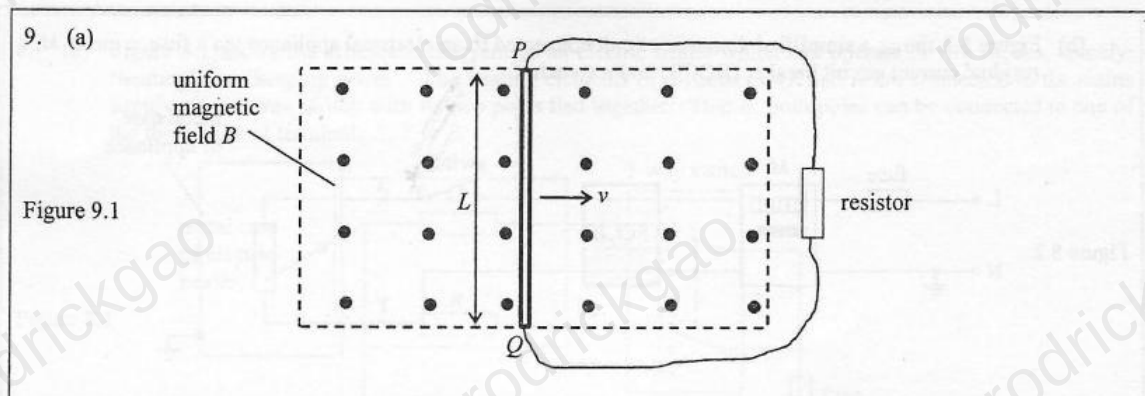
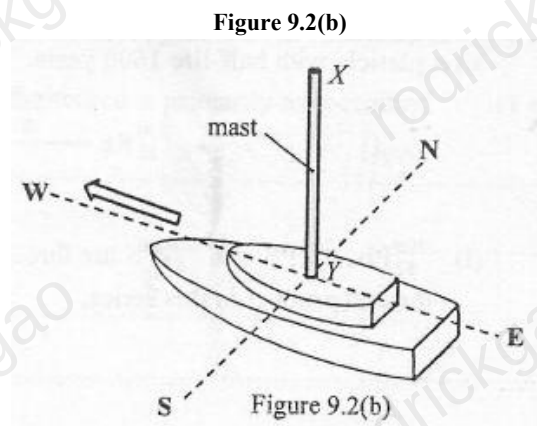
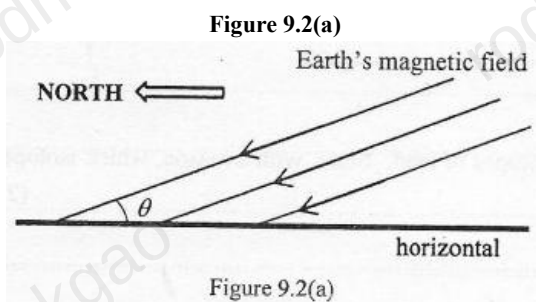


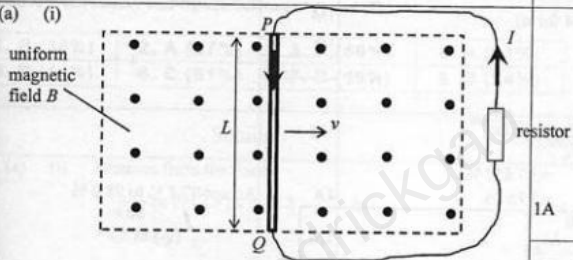
Figure 9.1

- (i) Indicate the direction of I in Figure 9.1. (1 mark)
 - (ii) Explain why an external force F is required to maintain the uniform motion of rod PQ. Find F in terms of the physical quantities given. (3 marks)
 - (iii) This set-up works as a generator. By considering the mechanical power input by external force F to the set-up, show that $\varepsilon = BLv$. (2 marks)
- (b) At a certain place the Earth's magnetic field runs along the S–N direction such that the field lines make an angle θ with the horizontal as shown in Figure 9.2(a). A ship with a vertical aluminium mast sails at sea along a straight course to the west as shown in Figure 9.2(b). As a result, an e.m.f. is induced across the mast XY.



- (i) Explain why it is only the horizontal component of the Earth's magnetic field that is cut by the mast which gives rise to this induced e.m.f. (1 mark)
- (ii) Given: length of mast $XY = 20 \text{ m}$; speed of the ship $= 6 \text{ m s}^{-1}$; Earth's magnetic field $= 50 \mu\text{T}$; $\theta = 30^\circ$. Referring to (a)(iii), calculate the e.m.f. induced across XY and state whether the distribution of free electrons along the mast is more at end X , more at end Y or uniform along XY . (3 marks)
- (iii) Suppose X and Y are connected by a cable running side-by-side with the mast so that they form a complete circuit. Explain whether there will be a current passing through it. (2 marks)

Official marking scheme

Solution	Marks	Remarks
9. (a) (i) 	1A	
(ii) By Lenz's law, a magnetic force F_B acts on the rod such that it opposes the rod's motion, an external force F is needed to balance F_B so as to maintain uniform motion (or constant v)	1A	Accept: Work done by an external force is needed to transfer mechanical energy into electrical energy.
$\therefore F = F_B = ILB$ (in magnitude)	1A	
Or $F \cdot v = I \xi$, $\therefore F = \frac{I \xi}{v}$	1A	
	3	
(iii) mechanical power input $= Fv$ $= (ILB)v$	1M	
power input = power output (electrical) $ILBv = I \xi$ $\xi = BLv$	1M	
	2	
(b) (i) Horizontal (component) is perpendicular to the mast. Or Vertical (component) is parallel to the mast.	1A 1A	
	1	
(ii) $\xi = (B \cos 30^\circ) L v$ $= (50 \times 10^{-6} \cos 30^\circ) (20) (6)$ $= 5.196152 \times 10^{-3} \text{ V}$ $\approx 5.20 \text{ mV}$ more electrons at end X	1M 1A 1A	by (a)(iii) using the horizontal component of B
	3	
(iii) No current. Both the cable and the mast cut the field lines in the same way, the e.m.f.'s produced are equal and thus oppose each other. Or No change in magnetic flux through the loop of the circuit.	1A 1A 1A	
	2	

(a) (i) Conventional current in the rod is from P to Q.

(ii) By Lenz's law the magnetic force opposes the rod's motion, so an external balancing force is needed; $F = BIL$.

(iii) Mechanical input power $Fv = BILv$ equals electrical output $I\xi$, hence $\xi = BLv$.

(b) (i) Only the horizontal field component is perpendicular to the vertical mast; the vertical component is parallel and is not cut.

(ii) $\varepsilon = (50 \times 10^{-6} \cos 30^\circ)(20)(6) = 5.20 \times 10^{-3} \text{ V} = 5.20 \text{ mV}$; more electrons collect at X.

(iii) The side-by-side cable develops an equal opposing motional e.m.f., so the net loop e.m.f. and current are zero.

Question 10

- (a) Part of the decay series of radium-226 (Ra-226) is shown below. Ra-226 decays to radon (Rn) by emitting an α particle with half-life 1600 years. The end product in the series is lead (Pb), which is stable.



Radium-226 decay series

- (i) ${}^{204}\text{Pb}$, ${}^{206}\text{Pb}$ and ${}^{208}\text{Pb}$ are three stable isotopes of lead. State, with a reason, which isotope can be the end product in this series. (2 marks)
- * (ii) In a certain laboratory, a Ra-226 source has been used for 50 years. Estimate the percentage of undecayed Ra-226 left after this period. (2 marks)
- (b) Spark counters can show the ionizing power of radiations. Figure 10.1 indicates the main features of a type of spark counter in school laboratories.

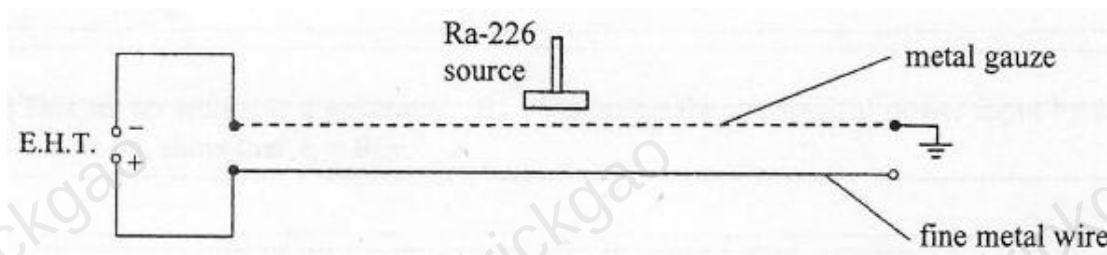


Figure 10.1

A spark counter consists of a fine metal wire mounted a few mm beneath an earthed metal gauze. The wire is connected to the positive terminal of an E.H.T. (Extra High Tension) supply so that a very intense electric field is set up between the wire and the metal gauze. When a Ra-226 source is brought near the gauze, sparks giving out flashes of light and crackling sound are produced at irregular intervals.

- (i) Explain why the sparks occur at irregular intervals. (1 mark)
- A Ra-226 source used in school laboratories is usually said to emit α , β as well as γ radiations.
- (ii) Explain why β radiation is also emitted even though the source is primarily an α -emitter. (1 mark)
- (iii) Why is the sparking mainly caused by α radiation rather than β or γ radiation? Suggest a simple way to verify this. (2 marks)

Official marking scheme

Solution		Marks	Remarks
10. (a)	(i) $226 - 206 = 20$ (multiple of 4 for α) $\therefore {}^{206}_{82}\text{Pb}$ is the end product	1M 1A	Accept 97.8 % to 98.0 %
		2	
	(ii) % undecayed Ra-226 left $\frac{N}{N_0} = e^{-\frac{\ln 2}{1600} \times 50}$	1M	
	$= 97.857\%$	1A	
	$\approx 97.9\%$		
	Or % undecayed Ra-226 left	1M	
	$= \left(\frac{1}{2}\right)^{\frac{50}{1600}}$	1A	
	$\approx 97.9\%$		
		2	
	(b) (i) $\therefore$ random process	1A	
	1		
(ii) Some of the daughter products of Ra-226 are also radioactive and may emit β particles.	1A		
	1		
(iii) Reason: weaker ionizing power of β and γ radiations	1A		
- raise the source to a distance greater than the range of α (a few cm) will cease to have sparks. - insert a paper between the source and the gauze, sparks will cease to produce. } Any ONE		1A	
		2	

(a) (i) Each α decay lowers the mass number by 4. Since $226 - 206 = 20$ is a multiple of 4, the possible stable end product is ${}^{206}\text{Pb}$.

(ii) Percentage remaining $= 100(1/2)^{(50/1600)} = 97.86\% \approx 97.9\%$.

(b) (i) Radioactive decay is random, so ionizing events and sparks occur irregularly.

(ii) Some radioactive daughter nuclides in the Ra-226 series emit β particles.

(iii) α radiation has much greater ionizing power than β or γ . Insert paper, or move the source beyond the short α range; the sparks should cease.